



Kurumsal
İş Sistemleri

e-Ledger Customer Web Service Document

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1.0.1	10.11.2014	Article 4	With accounting voucher data pattern Related Remarks added	Faruk Kaygısız
1.0.2	11.12.2014		Added general descriptions	Faruk Kaygısız
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1. e-Ledger General explanations

The e-Ledger system is a system created for keeping the books related to the accounting records that companies have to record in accordance with the legal obligation in electronic environment. The system currently covers the Journal Book and Kebir Book (Ledger). In the e- Ledger system, e-Ledger files are stored as XML files in XBRL standard. These files are signed with the companies' own financial seals. The signature value of these signed ledger files is added to a document called Berat, which contains a summary of company information, and this Berat file Income It must be sent to the system of the Administration. The Revenue Administration also adds its own signature to these Berat files and ensures the completion of the process.

Companies are obliged to keep their signed e-Ledger Documents in XBRL format and their Berat signed by RA and submit both documents to the authorities during official audits.

In Uyumsoft e-Ledger system, the creation of e-Ledger files in XBRL format is done automatically. e-Ledger Documents and Berat files will be signed with the companies' own financial seals. This signing process can be performed through Uyumsoft e-Invoice Portal.

In order to create e-Ledger files in XBRL format, Uyumsoft system expects data from companies in 2 different places.

1. Company information and accountant information must be entered on the company information screen in Uyumsoft Portal.
2. Sending the source data containing accounting receipt information in XML, CSV formats determined by Uyumsoft to the system via Web service or Portal interface.

Accounting receipts sent to the system are called source data. When loading source data If it is a format-compliant file, it is saved to the system and the status of this source data is "Completed".

If the status of the source data is Completed, it does not mean that the e-ledger process is completed. After the upload of the source data is completed, Uyumsoft Engine automatically creates the General Ledger(s) and Journal Ledger(s) in XBRL format depending on the source data.

During this creation process, the compliance of e-Ledgers with Schema/Schematron rules is also checked is performed. In addition, if the e-Ledger files exceed the specified file size (current size in the system is 20MB), it continues the creation process by dividing the ledgers into parts.

If one of the e-ledgers generated from the same data source fails to pass the error checks, the system cancels all e-ledgers generated from the same source.

If there is no error in the e-ledgers created based on the uploaded source file, e-Ledgers are created successfully and the unsigned file waits in the system in ready status.

The e-Ledgers are signed by the end user from the Uyumsoft portal interface using the company's financial seal.

Certificates of signed e-Ledgers are created and signed by the end user. The certificate is sent to the RA system by the end user.

All processes are completed after the signed versions of the certificates sent to the RA system are received into the system by Uyumsoft engine

2. Uyumsoft e-Ledger Customer Web Services

The Web Service application, which can be used by companies that want to create e-Ledgers to create e-Ledgers through Uyumsoft Independent Platform application and to monitor the status of the e-Ledgers created, is an intermediate platform that receives data from the customer environment and processes it in Uyumsoft e-Ledger Engine.

Web Service can be accessed from the following addresses:

Live System: <https://efatura.uyumsoft.com.tr/services/LedgerIntegration>

Test System: <https://efatura-test.uyumsoft.com.tr/services/LedgerIntegration>

(User Name for test system: Uyumsoft, Password:

Uyumsoft) Uyumsoft Portal can be accessed from the

following addresses Live: <http://portal.uyumsoft.com.tr/>

Test: <http://portal-test.uyumsoft.com.tr/>

(User Name for test portal: Uyumsoft, Password: Uyumsoft)

a. Return Value

All service methods return value in Response type. Response contains 3 types of fields.

IsSucceeded: This boolean field is *false* if there is an error in the operation, otherwise it is *true*.

Message: Usually used in error cases (*IsSucceeded = false*) to indicate what the error is.

Value: It refers to the field that returns the result of that operation according to each operation type.

3. Web Service Methods

i. UploadSource

Metod Signature	Response<StringResponse> UploadSource(LedgerSourceRequest)
Description	A source data (excel, csv, xml, xml in zip) to create e-Ledger It is the method that allows it to be transferred to Uyumsoft platform.
Parameters	1 LedgerSourceRequest object contains the information of the source data to be loaded. This object has 2 fields. - Data(byte[]) : Byte[] version of the created source file. - FileName: The name of the created Source file. This information should also include the file extension. The system can perform different operations according to the file extension.

Return Value	When the method with return type <code>Response<StringResponse></code> is called, Response.IsSuccessed = true if the operation succeeds, false if it fails. The unique ID of the source data generated in the Response.Value field will be returned in guid format in string type. The Response.Message field contains the reason for the error in case of an error will take it.
Sample Code (C#)	<pre> LedgeIntegrationClient client = CreateClient(); LedgeSourceRequest request = new LedgeSourceRequest(); byte[] zipfile = File.ReadAllBytes(@"C:\201401_ELedger.zip"); request.FileName = "201401_ELedger .zip"; request.Data = zipfile; StringResponse response = client.UploadSource(request); if (response.IsSuccessed==true) { MessageBox.Show(string.Format("The Source data you sent was created with Id {0}",response.Value)); } else { MessageBox.Show(string.Format("Error: {0} ", response.Message)); } </pre>

ii. GetSourceInformation

Metod Signature	<code>Response<ResponseOfLedgeSourceInformation></code> <code>GetSourceInformation(string identifier)</code>
Description	Status information about the sent source data depending on this source data It is the method where you can get the files and status etc. information of the created books and certificates.
Parameters	identifier Id field of the sent source data.

Return Value

In the method whose return type is `Response<ResponseOfLedgerSourceInformation>`, Value of Response returns the following values:

- `Value.Status`: Status of the sent Source file.
- `Value.StatusCode`: Status code of the sent Source file. (Listed below.)
- `Value.Logs`: The value of the log records related to the source file in list form.
- `Value.Ledgers`: The value in the form of a list of ledger files created based on the source file.
 - `Ledgers.Status`: Status of the ledger file.
 - `Ledgers.StatusCode`: Status code of the ledger file.
 - `Ledgers.Identifier`: Id of the ledger file (this Id must be used to call the ledger file from the system).
 - `Ledgers.DocumentId`: Ledger file Name. Legal document number in the format YEV2014..., KEB2014....
 - `Ledger.Logs`: Log records related to the ledger.

Source status codes: (`LedgerSourceStatus`)

Enum Value	Description	Code	Int Value
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	<table><tr><td>Waiting</td><td>Kuyrukta</td><td>0</td><td>0</td></tr><tr><td>Processing</td><td>İşleniyor</td><td>10</td><td>1</td></tr><tr><td>Completed</td><td>Tamamlandı</td><td>20</td><td>2</td></tr><tr><td>Error</td><td>Hata</td><td>30</td><td>3</td></tr><tr><td>Canceled</td><td>İptal</td><td>40</td><td>4</td></tr></table>	Waiting	Kuyrukta	0	0	Processing	İşleniyor	10	1	Completed	Tamamlandı	20	2	Error	Hata	30	3	Canceled	İptal	40	4																																
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Example Code(C#)	<pre>ResponseOfLedgerSourceInformation sourceInfoResponse = client.GetSourceInformation(sourceId); LedgerSourceStatus sourceStatus = sourceInfoResponse.Value.Status; //sent kaynak dosyanın durumu int sourceStatusCode = sourceInfoResponse.Value.StatusCode; //sent source status code of the file status code of the file LogRecordItem[] logs = sourceInfoResponse.Value.Logs; //logs of the source file sent. LedgerStatusWithLogInfo[] sourceInfo = sourceInfoResponse.Value.Ledgers; //information on the files (notebooks, certificates) related to the resource sent LedgerStatus ledgerStatus = sourceInfo[0].Status; //e-ledger status int ledgerStatusCode = sourceInfo[0].StatusCode; //e-ledger status code string ledgerIdentifier = sourceInfo[0].Identifier; //e-ledger Id string ledgerDocumentId = sourceInfo[0].DocumentId; //e-ledger file name in YEV2014... format name LogRecordItem[] ledgerLogs = sourceInfo[0].Logs; // e-ledger logs LedgerType ledgerType = ledgerData.Type; //type of ledger received</pre>																																																				

iii. GetLedgerData

Metod Signature	Response<ResponseOfLedgerData> GetLedgerData(string Identifier,bool CertificateFile)
Description	Returns the file of the given ledger Id.
Parameters	identifier The unique Id of the ledger file. This Id can be obtained from the GetSourceInformation method with the sourceId received when loading the source data. certificateFile: If you want to get the generated certificate of the specified ledger file, false should be passed if not true.

Return Value

In the method whose return type is `Response<ResponseOfLedgerData>`, the `Value` field of `Response` contains information about the ledger files.

- `Data`: the data of the ledger file in the form of a byte array.

- DocumentId: The name of the ledger file.
- FileName: The legal sequential name of the ledger file. If the ledger record is divided into more than one file, it takes the extension _01,_02
- FileType: Gives the type of the ledger file (signed ledger, unsigned ledger and certificate)
- Status: Ledger status
- StatusCode: Ledger status code
- Type: Information on whether it is Journal or Kebir ledger.

Ledger status codes: (LedgerStatus)

Enum Value	Description	Code	Int Value
Building	Hazırlanıyor	5	1
FileSigned	İmzalandı	20	3
BeratSignedByCustomer	Berat İmzalandı	40	5
Completed	Tamamlandı	100	7
Error	Hata	300	9

File Types (LedgerFileTypes)

Enum Value	Description	Int Value
SignedLedgerFile	İmzalı Defter	1
SignedCertificateFile	İmzalı Berat	3

Ledger Types (LedgerType)

Enum Value	Description	Int Value
Journal	Yevmiye Defteri	0
Ledger	Kebir Defteri	1

Example Code(C#)

```
ResponseOfLedgerData ledgerDataResponse = client.GetLedgerData(ledgerId, false);
LedgerData ledgerData = ledgerDataResponse.Value;
byte[] file = ledgerData.Data; //Ledger file
string documentId = ledgerData.DocumentId; //ledger file name YEV2014... formatındaki ad
string fileName = ledgerData.FileName; //Legal file name of the ledger file created by Uyumsoft.
LedgerFileType fileType = ledgerData.FileType; //type of file received
LedgerStatus status = ledgerData.Status; //status of the ledger or certificate file
int statusCode = ledgerData.StatusCode; //status code of the ledger or certificate file
LedgerType ledgerType = ledgerData.Type; //type of ledger received
```

iv. CancelLedgerSource

Metod Signature	Response<bool> CancelLedgerSource(string Identifier)
Description	It is used to cancel the given source file Id.
Parameters	identifier:Source data sent by the UploadSource method is the name of the file.
Return Value	The method whose return type is Response<bool> also returns true or true in the Value field of Response. also returns false. If the cancellation operation is successful, true will return false if it fails.
Sample Code(C#)	<pre>FlagResponse cancelSourceResponse = client.CancelLedgerSource(sourceId); bool cancellationStatus = cancelSourceResponse.Value;</pre>

v. CancelLedger

Metod Signature	Response<bool> CancelLedger (string Identifier)
Description	It is used to cancel the given ledger Id.
Parameters	identifier The unique Id of the ledger to be cancelled.
Return Value	The method whose return type is Response<bool> also returns true or true in the Value field of Response. also returns false. If the cancellation operation is successful, true will return false if it fails.
Sample Code(C#)	<pre>FlagResponse cancelLedgerResponse = client.CancelLedger(ledgerId); bool cancellationStatus = cancelLedgerResponse.Value;</pre>

4. Accounting Receipt data structure

The fields that should be included in the XML, CSV, etc. documents to be sent to Uyumsoft independent platform in order to create e-Ledger files and the explanations of these fields are as follows:

	Domain Name	Domain Name	Required Field	Field Description
Entry Header - Journal Header Information	StartDate	Yevmiye Dönemi Başlangıç Tarihi	Yes	Journal Period Start Date will be entered
	EndDate	Yevmiye Dönemi Bitiş Tarihi	Yes	End Date of Journal Period will be written
	EnteredBy	Yevmiye Maddesini Kaydeden Kişi	Yes	The name of the person who recorded the Journal Entry will be written.
	EnteredDate	Yevmiye Kayıt Tarihi	Yes	Journal Entry Date will be written.
	EntryNumber	Yevmiye Fiş Numarası	Yes	Accounting receipt number will be written.
	EntryComment	Yevmiye Maddesi Açıklaması		Journal entry description will be written.
	BatchID	Şube ID		e-Ledger A branch with permission to create Ledgers on behalf of the branch branch code with 4 digits must be entered in this field.

				e-Ledger A branch with permission to create Ledgers on behalf of the branch Branch name should be entered as "X BRANCH" in this field.
	BatchDescription	Şube Kodu		
	TotalDebit	Toplam Borç	Evet	Total amount of debt will be written in the journal item.
	TotalCredit	Toplam Alacak	Evet	Total amount of receivable will be written in the journal entry.
	DocumentType	Belge Türü	*	If the journal entry is based on a document, the document type must be entered in this field. Allowed document type names (check: Check, invoice: Invoice, order-customer: Customer Order Document, order-vendor: Vendor Order Document, voucher: Promissory note, shipment: Freight, receipt: Receipt, other: Other)
	DocumentTypeDescription	Belge Tipi Açıklaması	*	Other in the field Document Type If entered, the description of this document must be entered in this field.
	DocumentNumber	Belge Numarası	*	If the Document Type is Invoice, Check or Other, you can enter the number of the document in this field. must be entered.
	DocumentDate	Belge Tarihi	*	If the Document Type is Invoice, Check or Other, the date of the document must be entered in this field.
	PaymentMethod	Ödeme Yöntemi		If there is a payment in the journal item, in the payment method field should be indicated as free text. This field will continue in each accounting period, starting from 1 and increasing by 1 in each journal entry. Number in this field should not be skipped and should come sequentially in the XML or CSV form sent. This field
Entry Detail - Journal Detail Information	EntryNumberCounter	Yevmiye Madde Numarası		When it is not created, Uyumsoft system will automatically give journal item number.
	AccountAccountMainID	Ana Hesap Kodu	Evet	Main Account code as 3 digits will be entered (100,120 etc.)
	AccountAccountMainDescription	Ana Hesap Açıklaması	Evet	Main Account description in this field must be entered (CASH, BANKS etc.)

AccountID	AccountSub	Alt Hesap Kodu		Subaccount code will be entered in this field (100.01, 120.01 etc.). Sub-account the beginning of the subject and the beginning of the Main Account code must be the same (like 100, 100.01)
AccountDescription	AccountSub	Alt Hesap Açıklaması		Subaccount description in this field should be entered (TL CASH, BANK ZIRAAT etc.)
Amount		Miktar	Evet	The monetary amount related to the line of the journal entry will be written.
DebitCode	CreditCode	Borç ya da Alacak Kodu	Evet	Journal Line will be C if it is related to Receivable and D if it is related to Debit. (C: Credit, D: Debit)
DetailComment		Yevmiye Detay Açıklaması		Description of journal line will be written.
PostingDate		Yevmiye Kayıt Tarihi		Must be equal to EnteredDate
LineNumber	Counter	Yevmiye Satır Numarası		This field will continue in each accounting period, starting from 1 and increasing by 1 in each journal line. It will continue from where it left off in the next journal entry. No number skipping should be done in this field and it should come sequentially in the XML or CSV form sent. When this field is not created, the Uyumsoft system automatically creates the journal will give a line number.
VknTckn		Defteri Gönderen Firma VKN ya da TCKN numarası		Working with more than one company This is a control field created to prevent companies from accidentally uploading a ledger on behalf of a different VKN.

